

AS • Edexcel • Physics

 32 mins  3 questions

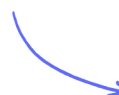
Structured Questions

Forces & Momentum

Force & Acceleration / Mass, Weight & Gravitational Field Strength / Core Practical 1: Investigating the Acceleration of Freefall / Newton's Third Law of Motion / Momentum / Conservation of Linear Momentum

Medium (2 questions)	/20
Hard (1 question)	/12
Total Marks	/32

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Medium Questions

- 1 (a)** A student released a ping pong ball in front of a metre rule and used a phone camera to record the motion of the ball as it fell. The phone camera captures 60 images per second, which may be played back one image at a time.

The ball was dropped from a height such that it reached its terminal velocity as it passed the metre rule.

- i) Explain how the terminal velocity of the ball could be determined using the phone camera recording.

(4)

- ii) Explain how a systematic error could affect the value obtained for the terminal velocity.

(2)

(6 marks)

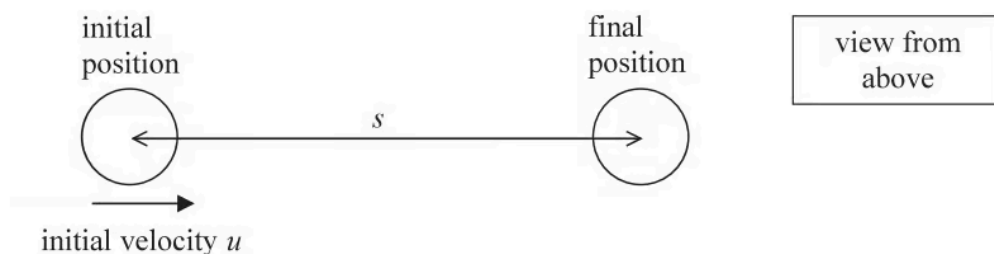
- (b)** This experiment could have been attempted using a stopwatch to measure the time as the ping pong ball fell.

Explain an advantage of using a phone camera rather than a stopwatch.

(2 marks)

2 (a) A student carried out an experiment with coins.

She gave a 2p coin a sharp tap, so that it slid along a horizontal surface and came to rest as shown.



The student recorded the distance s moved by the coin.

She then replaced the 2p coin with a 1p coin and repeated the process.

The student read that the frictional force between an object and a surface is directly proportional to the mass of the object. She suggested that, in her experiment, u is directly proportional to $\sqrt{s}$ and is independent of the mass of the coin.

Discuss the validity of this suggestion.

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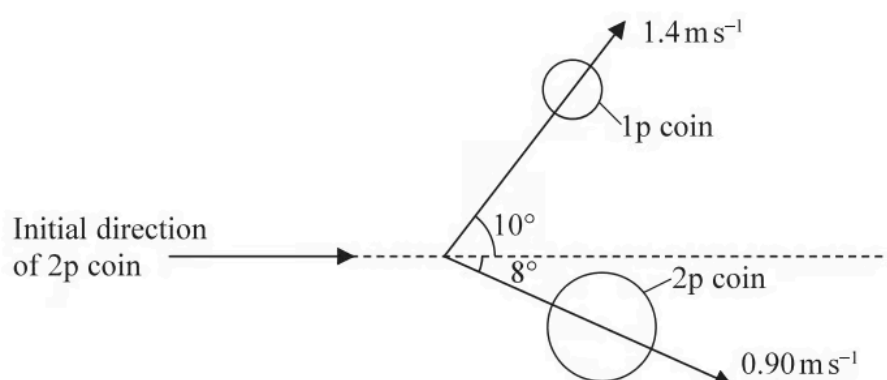
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(6 marks)

(b) She arranged a collision between a 2p coin and a stationary 1p coin. She noted the

directions in which the coins moved after the collision and determined their velocities.



i) Show that the velocity of the 2p coin just before the collision was about 2 ms^{-1} .

mass of 2p coin = 7.1 g

mass of 1p coin = 3.6 g

(4)

ii) Show that the collision was inelastic

(2)

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(6 marks)

Hard Questions

- 1 (a) In the sport of curling, two teams of 'curlers' take turns sliding polished granite stones across an ice surface towards a circular target marked on the ice.



commons.wikimedia.org

A stone of mass 19.6 kg is accelerated uniformly for 1.25 s before being released by a curler.

The stone then decelerates uniformly to rest, travelling 32.5 m in a time of 17.5 s.

Calculate the average useful power developed by the curler in accelerating the stone.

Average power =

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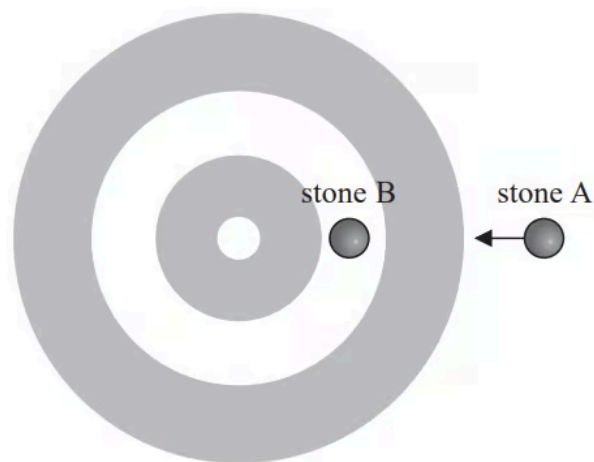
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(4 marks)

- (b) Stone B is stationary. Stone A travels towards the target and makes a direct hit on stone

B as shown. Both stones have mass m .



The collision is elastic. Just before the collision stone A has a velocity v . After the collision stone B moves off with velocity v .

Discuss how the relevant conservation laws apply to this collision.

(6 marks)

- (c) While a stone is moving towards the target, the curlers vigorously sweep the ice directly in front of the stone.

Explain why this may make the stone travel further.

(2 marks)